Issue	**1**
Factory ID	**25797**
Date of Issue	**13-Apr-2026**

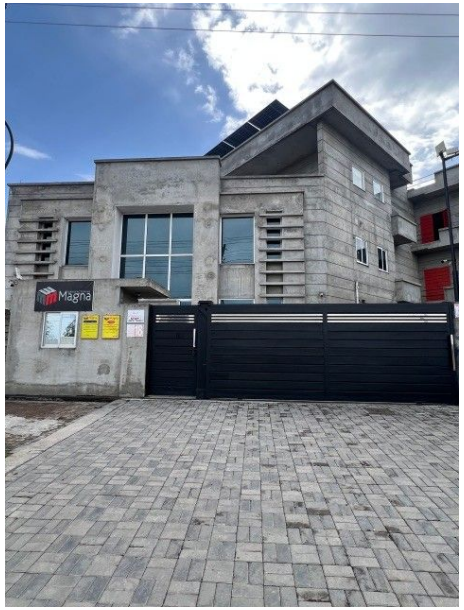
Magna Manufacturing (New Site)

13 Km Daska Road
Sialkot
Punjab Pakistan

Latitude: [32.39948], longitude: [74.44158]
Date of inspection: 17-Mar-2026

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 17-03-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

Contents

1. Executive Summary 5
2. Building Extents 7
   - 2.1 Building Use 9
   - 2.2 Utility Source 10
   - 2.3 Backup source 11
3. Documentation 12
4. Electrical Safety Measures 13
5. Remediation Support 14
   - 5.1 Engineering Support 14
6. Diagrams & Drawings 15
   - 6.1 Sub-station room layout drg 15
   - 6.2 Single Line Diagram (SLD) 16
   - 6.3 Electrical layout drawing 17
7. DBs & Elec Protection Systems 18
   - 7.1 Circuit breakers 18
   - 7.2 Hot Spots 21
   - 7.3 Earthing system 25
   - 7.4 Panel/cabling/busbar 29
   - 7.5 Loop Connection 32
   - 7.6 Multiple Connections 34
   - 7.7 Openings 36
   - 7.8 Adequate accessibility 38

7.9 Lightning Protection System (LPS) 40
8. Fire Alarm & Detection System (FADS) 42
8.1 Non-functional or incomplete 42
9. Emergency Lighting 45
9.1 Provided but non-functional 45
9.2 Provided but insufficient 47
9.3 Independent power source 50
10. Generator, Transformer & Sub-Station Room 51
10.1 Battery 51
11. Cabling & Wiring 53
11.1 Cable support and protection 53
11.2 Splicing of cables 55
11.3 Colour coding 58
12. Assessment Limitations and Assumptions 60

1. Executive Summary

On 17-Mar-2026, Azka Riaz and Muhammad Bilal Butt on behalf of the Pakistan Accord carried out an initial electrical inspection for Magna Manufacturing (New Site), a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with Humza Mahmood (Director), Umar Mahmood (Director) and other staff members of Magna Manufacturing (New Site). The assessment was conducted as part of an overall Structural, Fire, and Electrical initial inspection.
The factory operates in single shift. Approximately 47 workers operate in the factory with 1 worker being female.
The factory consists of one main building and two ancillary structures owned by Magna Manufacturing (New Site). During the inspection, the areas highlighted in Figure 1.1 were surveyed, with the site boundary indicated in red dashed line.

- **Main Building** is a four storey (Ground + 3) building, used for cutting, storage, shipment, raw material inspection, sampling, offices, (Ground Floor), sewing, packing, checking, offices (First Floor), dining and prayer area (Second Floor), solar panels and overhead water tank (Third Floor). The third floor of the main building is currently vacant but planned for future production use.
- **Electrical Room** is a single storey (Ground only) structure, used for generator operations (Ground Floor).
- **Gate Office** is a single storey (Ground only) structure, used for office (Ground Floor).

A summary of the main concerns in relation of life safety systems identified during the inspection (non-exhaustive) that should be actioned are as follows:

- **Fire Alarm and Detection System (FADS):** The FADS across the facility was found to be conventional and deficient, with the use of non-addressable and non-listed components, inadequate detection coverage, and non-fire-rated cabling. These shortcomings significantly compromise system reliability and fire detection capability, posing serious life safety risks and increasing the potential for injury, fatalities, property damage, and operational and reputational losses in the event of a fire.
- **Abnormal Temperature:** Abnormal temperatures were recorded within electrical panels in various locations. These hotspots pose a risk of equipment failure, fire hazards, and system downtime. It is recommended that the factory conduct a detailed assessment to identify the root cause and implement corrective actions as necessary.
- **Earthing system:** The earthing system was found to be inadequate across the facility. Earthing pits were inaccessible, most distribution boards lacked earthing conductors, and none of the electrical machines were provided with earth connections. The absence of a proper earthing system poses significant safety risks, including electric shock, fire hazards, and potential equipment damage.
- **Emergency lighting System:** The emergency lighting system exhibited several deficiencies, including inadequate coverage along egress routes, aisles and emergency stairs, and lack of functionality, which hinders safe evacuation during power outages and increases accident risks.

We see no reason to suspend operations in the facility due to these concerns, subject to the actions identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.1. Satellite view of the factory– Red dashed line indicates site boundary

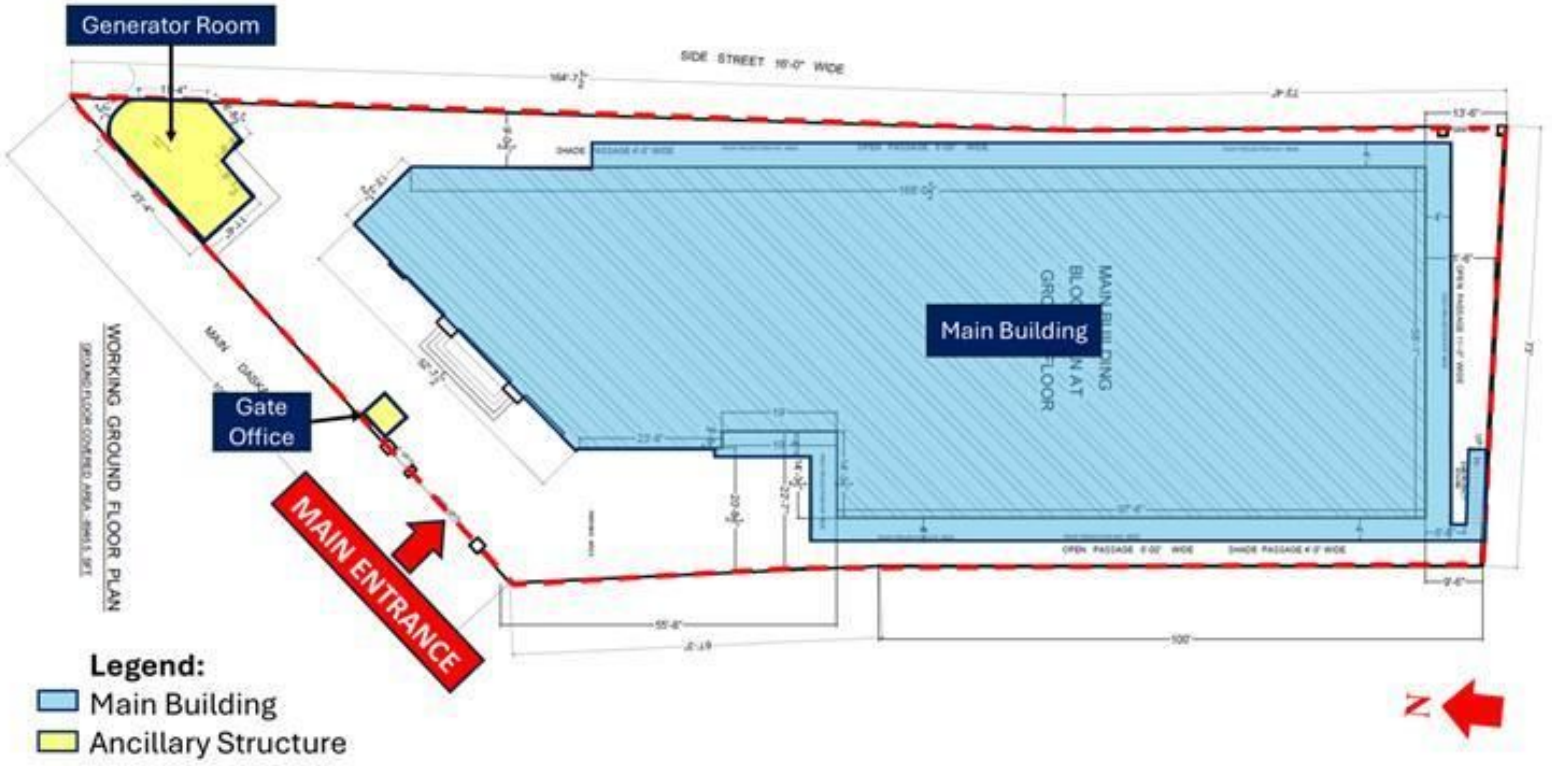


Figure 1.2. Site layout plan - Red dashed line indicates site boundary

2.1 Building Use

The factory comprises one main building and two ancillary structures. The building use per floor is as follows:

1. Main Building

- L00-Ground Floor: Cutting, Storage, Shipment, Raw Material Inspection, Sampling, Offices
- L01-First Floor: Sewing, Packing, Checking, Offices
- L02-Second Floor: Dining Area, Prayer Area
- L03-Third Floor: Solar Panels, Overhead Water Tank, Vacant Shed (Planned for Future Production Use)

2. Electrical Room

- L00-Ground Floor: Electrical panels, portable generator

3. Gate Office

- L00-Ground Floor: Security Office and Fire Alarm Control Panel

2.2 Utility Source

The facility receives a 440 V connection from Gujranwala Electric Power Company (GEPCO) as its primary power source, supplied through a GEPCO-owned 100 kVA step-down transformer installed near the main gate. The incoming power is routed to the Low Tension (LT) panel located in the electrical room, from where it is distributed throughout the facility via underground cables to various distribution boards. The facility also has an 8 kVA hybrid and a 30 kVA on-grid solar power system installed on the rooftop.

2.3 Backup source

The facility has one low-voltage petrol-powered 7 kVA generator located in the electrical room, serving as a backup power source. The fire alarm and emergency lighting systems are provided with built-in battery backup.

3. Documentation

The table shows the details of the documentation provided by the factory:

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical Layout Drawing	Yes	Unknown	Document does not match site conditions and is not up to date as per the required standards.
2	Single Line Diagrams	No		
3	Fire Alarm Drawing	Yes	Unknown	Document does not match site conditions and is not up to date as per the required standards.
4	Earth Test Report	No		
5	Thermographic Test Report	No		
6	Insulation Resistance Test Report	No		
7	LPS Risk Assessments	No		
8	Periodic Maintenance Record	No		
9	Transformer Oil Testing Report	N/A		

4. Electrical Safety Measures

During our electrical site inspection of Magna Manufacturing (New Site), a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within Magna Manufacturing (New Site). Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a licensed professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202603-064423

Timeline	Remediation Plan
Medium Term CAP-0075908	Factories should begin remediation of CAP items—such as cleaning, cable organization, and basic corrections—using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Design Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no substation room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202603-064404

Timeline	Remediation Plan
Medium Term CAP-0075911	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0075912	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were missing.

This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202603-064403

Timeline	Remediation Plan
Medium Term CAP-0075909	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0075910	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the drawings available with the factory lacked essential electrical layout drawings, including small power layouts, lighting layouts, and containment layouts. Furthermore, the available drawings were not up to standard and did not reflect actual site conditions.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202603-064405

Timeline	Remediation Plan
Medium Term CAP-0075913	Update electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0075914	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.

During the survey the circuit breakers installed throughout the facility inside the electrical panels and distribution boards were inspected.

i. It was observed that the circuit breaker is improperly mounted without secure fixing or proper enclosure, with exposed and loosely routed cables through inadequately terminated flexible conduits and unsealed entries. This condition may lead to loose connections, overheating, arcing, short circuits, and increased risk of electric shock due to exposure of live parts and mechanical stress on terminations. The issue was noted at LT room.

ii. It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes.
a) LT Panel: 35 sq. mm cable connected to a 200A circuit breaker.
b) Storage Area DB (Ground Floor): 4 sq. mm cable connected to a 100A circuit breaker.
c) Corridor DB (Ground Floor): 4 sq. mm cable connected to a 100A circuit breaker.
d) Office DB (First Floor): 4 sq. mm cable connected to a 60A circuit breaker.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202603-064408

Timeline	Remediation Plan
Medium Term CAP-0076103	i. Securely mount the circuit breaker in a proper enclosure with adequate IP rating, ensuring all cable entries are gland-sealed, conduits are properly terminated and supported, and no conductors remain exposed.

Medium Term CAP-0076104	ii. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - A 35 mm² cable was connected to a 200 A circuit breaker at the LT Panel



Figure 2.2 - A 4 mm² cable was connected to a 100 A circuit breaker at the Storage Area DB (Ground Floor)





Figure 2.3 - A 4 mm² cable was connected to a 100 A circuit breaker at the Corridor DB (Ground Floor)



Figure 2.4 - A 4 mm² cable was connected to a 60 A circuit breaker at the Office DB (First Floor)

Figure 2.5 - Improperly mounted circuit breaker with exposed and poorly terminated cables at LT room

7.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey identified numerous hot spots within panels and distribution boards. Hot spots in electrical distribution systems can be caused by several factors, including:

> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation

It was also found that the thermographic report was not available at the site. Without thermography testing, faults such as overheating components or loose connections may go unnoticed, which increases the risk of fire. Early detection through thermography can prevent fire incidents.

The categories of identified hot spots have been evaluated as per the ANSI NETA MTS 2019 standard, Table 100.18. Hot spots were observed, particularly at the following locations:
i. 73°C hotspot found at the LT panel.
ii. 84.8°C recorded in the first floor office area.
iii. 70.8°C recorded in the cutting machine area.
iv. Elevated temperature observed in the first floor area at the solar DB (hybrid inverter).

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature

event that can cause severe injuries or fatalities to personnel working near the affected equipment.F-202603-064414

Timeline	Remediation Plan
Immediate CAP-0076107	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.
Medium Term CAP-0076108	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0076109	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating. Submit the thermographic test report to the Pakistan Accord for review.



Figure 2.6 - Hot spot found on circuit breaker first floor office area (thermal image)



Figure 2.7 - Hot spot found on circuit breaker first floor office area (digital image)



Figure 2.8 - Hot spot found on cable termination at LT panel (thermal image)



Figure 2.9 - Hot spot found on cable termination at LT panel (digital image)



Figure 2.10 - Hot spot found on cable termination cutting machine area (thermal image)



Figure 2.11 - Hot spot found on cable termination cutting machine area (digital image)

7.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. Comprehensive drawings for the earthing system were absent. The lack of proper documentation and identification of the earthing and neutral system components pose safety risks.
ii. The earth pits were inaccessible and unsuitably located outside the factory.
iii. The earthing cables were not appropriately sized according to the relevant earthing standards at LT panel, LT room.
iv. The earthing system was absent in multiple locations. Several distribution boards lacked earthing cables, and many electrical machines did not have proper earth connections. The issue was noted at storage area, water cooler, stitching machines, cable tray and metal fixings.
v. Equipotential bonding for the cable trays and metallic panel covers was missing at distribution boards and cable tray installed on first floor.
vi. It was found that the earth testing record was not available at the site. Without earth testing records, it is difficult to ensure that the grounding system is functioning correctly and that maintenance activities are being properly conducted.

The lack of proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202603-064415

Timeline	Remediation Plan
Medium Term CAP-0076118	i. Design and evaluate the earthing system installation throughout the facility by preparing a comprehensive earthing system diagram, including identification of all earthing and neutral system components. Ensure compliance with BS 7671: 18th Edition + A2 (2022), Regulation 514.5.1 regarding proper identification and labeling and, submit it to Pakistan Accord for review.
Medium Term CAP-0076119	ii. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54.
Medium Term CAP-0076120	iii. Replace all earthing cables with appropriately sized cables as per the requirements of BS 7671 18th Edition + A2 (2022) Appendix 4, Section 5. Verify that all earthing conductors meet the minimum cross-sectional area specified.

Medium Term CAP-0076121	iv. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.
Medium Term CAP-0076122	v. Provide equipotential bonding for all cable trays, metallic panel covers, and other exposed conductive parts, complying with BS 7671: 18th Edition + A2 (2022).
Medium Term CAP-0076123	vi. Prepare and submit the earth testing record to the Pakistan Accord for review.
Longer Term CAP-0076124	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.12 - Earthing cable at the LT panel found to be undersized, contrary to applicable earthing standards



Figure 2.13 - Disconnected earth cable observed in storage area DB ground floor



Figure 2.14 - Earth connection found to be missing at water cooler



Figure 2.15 - Earthing system absent at first floor checking and packing area distribution boar



Figure 2.16 - No earth connection found for first floor stitching machines



Figure 2.17 - Cable tray not bonded with earth at first floor



Figure 2.18 - Inaccessible earth pit outside factory wall

7.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables and busbars throughout the facility.

i. In a number of external areas it was observed that electrical components were not suitable for the environmental conditions in which they were installed. The issue was noted at backside gallery area.
ii. Electrical panel exhibited signs of corrosion, indicating potential exposure to moisture or other corrosive elements. The issue was noted at backside gallery area.
iii. Inadequate cable dressing was observed in most of the distribution boards, with conductors not properly arranged, secured, or terminated neatly. This resulted in untidy wiring, poor segregation between circuits, and unnecessary strain on terminations, which can lead to loose connections, overheating, and difficulty during inspection or fault-finding. The issue was noted at all distribution boards installed at factory.
iv. Phase separators were missing in some panels posing the risk of accidental contact between phases. The issue was noted at ground floor and first floor distribution boards.

The utilisation of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202603-064416

Timeline	Remediation Plan
Medium Term CAP-0076009	i. Identify and replace electrical panels, cables, and busbars that are not suitable for the environmental conditions in which they are installed in accordance with Section 512.2 of BS 7671 18th Edition + A2 (2022).
Medium Term CAP-0076010	ii. Replace or refurbish corroded electrical panels to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials.
Medium Term CAP-0076011	iii. Ensure all distribution boards are properly dressed and organized to prevent mechanical stress on terminals, in accordance with BS 7671:2018+A2:2022.
Medium Term CAP-0076012	iv. Install missing phase separators between incoming and outgoing terminations to provide adequate insulation and prevent flashovers between phases, in accordance with BS 7671:2018+A2:2022.

Longer Term CAP-0076013	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.19 - Missing phase separators and inadequate cable dressing observed at LT panel electrical room





Figure 2.20 - Faulty monitoring meters at ground floor DB



Figure 2.21 - Missing phase separators and inadequate cable dressing observed at first floor DB



Figure 2.23 - Non IP rated electrical component installed at backside gallery area

Figure 2.22 - Corrosion DB found at backside gallery area

7.5 Loop Connection

This relates to the entire facility.

During the inspection loop connections were identified within the electrical panels.

In these loop connections, one circuit breaker is energised by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system. The issue was noted at LT panel and change over panel at LT room and storage area DB ground floor.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202603-064409

Timeline	Remediation Plan
Medium Term CAP-0076105	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilized to provide each circuit with the required live, neutral, and earth connections.



Figure 2.24 - Loop connection at storage area distribution board ground floor

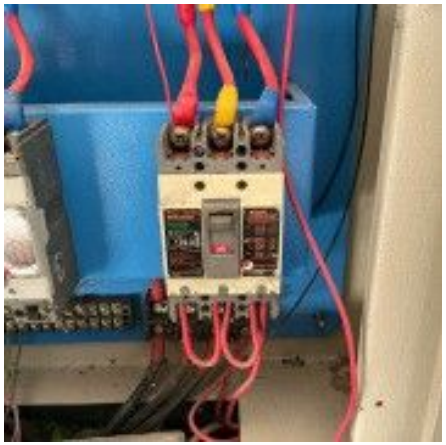


Figure 2.25 - Loop connection at LT room change over panel



Figure 2.26 - Loop connection at LT panel

7.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility.

In these cases, multiple circuits were being protected by a single circuit breaker. The issue was noted at LT panels, storage area DB ground floor, office DB first floor.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202603-064407

Timeline	Remediation Plan
Medium Term CAP-0076102	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022).



Figure 2.27 - Multilooping observed at ground floor storage area distribution board

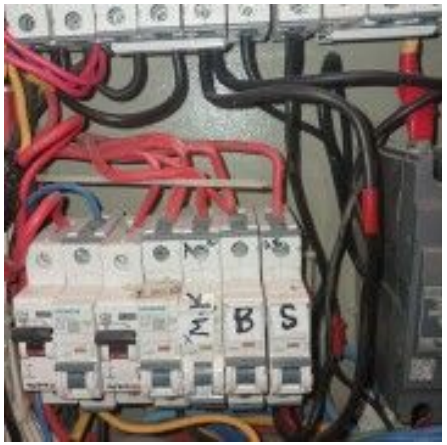


Figure 2.28 - Multilooping observed at first floor office area distribution board



Figure 2.29 - Multilooping observed at LT panel

7.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

Unsealed openings were discovered in the panels and distribution boards at electrical room, on-grid solar DB outside main building and busbar DB.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202603-064418

Timeline	Remediation Plan
Medium Term CAP-0076003	Corrective actions are required to seal all openings and replace missing panels to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0076004	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.30 - Openings found in LT panel and electrical room



Figure 2.31 - Openings found in on-grid PV solar system DB



Figure 2.32 - Openings found in change-over panel electrical room

7.8 Adequate accessibility

This relates to the entire facility.

During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

It was found that the electrical panels are installed at elevated positions and in areas that are not easily accessible, particularly the on-grid solar DB near the busbar panel outside the main building and the solar DBs located at the backside gallery.

The excessive mounting height and surrounding obstructions significantly limit safe and convenient access for inspection, maintenance, and emergency intervention. Such restricted accessibility can hinder routine servicing, delay fault diagnosis, and increase response time during electrical emergencies.

These conditions may compromise the operational reliability and safety of the electrical system, as proper maintenance and prompt corrective actions are difficult to perform.F-202603-064412

Timeline	Remediation Plan
Medium Term CAP-0076106	Electrical equipment shall be installed with adequate clearances to ensure safe access for inspection, operation, maintenance, and replacement of components, in accordance with Section 132.12 of BS 7671:2018 + A2 (2022).



Figure 2.33 - Inadequate accessibility for on-grid solar DB near the busbar panel outside the main building

7.9 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

A lightning protection system was provided, however, there was no study available to confirm its adequacy.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202603-064417

Timeline	Remediation Plan
Medium Term CAP-0076006	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment as per BS EN 62305 (Protection against Lightning). This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0076007	If the risk assessment concludes that an LPS is necessary then hire a qualified engineer to perform an adequacy study of the existing LPS. This study should evaluate whether the current system meets the necessary design and performance standards. If the study finds that the current LPS is inadequate, make the necessary upgrades to bring the system up to standard such as BS EN 62305 (Protection against Lightning).



Figure 2.34 - Lightning arrester installed on roof top of the main building

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

A conventional fire alarm system was installed at the facility. During the inspection, several deficiencies were noted, including the use of non-addressable and non-listed equipment, inadequate detection coverage, and the use of non-fire-rated cables, all of which pose a significant life-safety risk during a fire emergency.

A non-functional or incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm. A delayed response during a fire event can significantly increase the impact on personnel, goods, and property.F-202603-064420

Timeline	Remediation Plan
Medium Term CAP-0075990	i. Submit design documentation for the new FADS system to Pakistan Accord for review prior to procurement of any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Medium Term CAP-0075991	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier.
Longer Term CAP-0075992	iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.
Longer Term CAP-0075993	iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.

Longer Term CAP-0075994	v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.
Longer Term CAP-0075995	vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.35 - No FADS installed at electrical room



Figure 2.36 - Non fire rated cables used for FADS in the facility



Figure 2.37 - Fire alarm control panel installed at gate office building



Figure 2.39 - No FADS installed in third floor shed



Figure 2.38 - No FADS installed in third floor main buildings



Figure 2.40 - Inside view of fire alarm control panel installed at gate office building

9. Emergency Lighting

9.1 Provided but non-functional

The section refers to the entire facility.

During the inspection it was observed that emergency exit sign was provided but not functional at first floor main building.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202603-064413

Timeline	Remediation Plan
Medium Term CAP-0075996	Conduct a thorough inspection of all emergency exit lights throughout the facility to identify specific locations and units that are non-functional. This includes checking the lights, power sources, batteries, and wiring connections. Repair any faulty emergency exit lights immediately. If repair is not feasible, replace the defective units with new, compliant emergency exit lighting fixtures to ensure adequate illumination for safe evacuation during an emergency.
Longer Term CAP-0075997	Establish a regular maintenance and testing schedule to routinely check the functionality of all emergency lights and emergency exit lights. This should include periodic inspections, tests, and prompt corrective actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the proper procedures for inspecting and maintaining emergency lighting and exit systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.41 - Non functional emergency exit light at first floor main building

9.2 Provided but insufficient

The section refers to the entire facility.

During the inspection the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. Several areas lack sufficient emergency lighting coverage. The issue was noted at stairs, backside gallery area and electrical room.
ii. External emergency lighting was not provided, preventing people from reaching a place of safety outside the facility during nighttime evacuations. The issue was noted at assembly area.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202603-064421

Timeline	Remediation Plan
Medium Term CAP-0076000	i. Prepare a detailed emergency lighting design based on a comprehensive site survey to identify areas with inadequate coverage, in accordance with the Building Code of Pakistan – Fire Safety Provisions 2016. Submit the proposed emergency lighting layout to Pakistan Accord for review and approval. Upon approval, install additional emergency luminaires in the identified areas to ensure adequate illumination for safe evacuation.
Medium Term CAP-0076001	ii. Install external emergency luminaires to enable safe evacuation during hours of darkness all the way to a safe place outside the building(s).
Longer Term CAP-0076002	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.42 - Backside gallery area with no emergency lights installed



Figure 2.43 - Emergency lights and illuminated exit signs not provided for first and second floor stairs



Figure 2.44 - No emergency light provided for the electrical room



Figure 2.45 - No emergency light provided at assembly area



Figure 2.46 - No emergency light on stairs in the factory

9.3 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. Further the emergency lights and exit signs should be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system. According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202603-064419

Timeline	Remediation Plan
Medium Term CAP-0075998	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately upon failure of general lighting power, and provide an independent circuit for the emergency lighting system, as required by the Building Code of Pakistan Fire Safety Provisions 2016. Submit circuitry design diagrams to Pakistan Accord for review.
Longer Term CAP-0075999	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Provide independent power circuit to emergency lights. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

10. Generator, Transformer & Sub-Station Room

10.1 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

The battery terminals were exposed and not properly insulated at electrical room, increasing the risk of electrical shock or short circuits.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202603-064422

Timeline	Remediation Plan
Medium Term CAP-0075915	Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Longer Term CAP-0075916	Establish a routine inspection and maintenance protocol to regularly check and maintain proper insulation of all battery terminals.



Figure 2.47 - Battery terminals found unprotected at the 7 kVA petrol-powered generator in the electrical room

11. Cabling & Wiring

11.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

Improper or inadequate cable tray/duct support was observed at the stitching area, 1st floor. Weak or incorrectly installed supports compromise the integrity of the containment system, increasing the likelihood of sagging or cable damage.F-202603-064406

Timeline	Remediation Plan
Medium Term CAP-0075919	Ensure all cable trays and ducts are properly supported and structurally sound. Replace or reinforce any weak, unstable, or inadequately mounted systems to maintain cable integrity and system safety. (Reference: BS 7671 18th Edition + A2 (2022), Regulation 522.8.5)
Longer Term CAP-0075920	Establish a routine maintenance schedule to assess the condition and support of cable trays. Ensure proper reinforcement and alignment to prevent future sagging or instability. Document findings and corrective actions in maintenance logs.



Figure 2.48 - Inadequate support for cable duct at stitching area first floor

11.2 Splicing of cables

This relates to the whole facility.

During the inspection the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. Cables terminations within panels and switchgears were made without cable lugs, resulting in breaking strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. The issue was noted at Storage Area DB (Ground Floor) and Office DB (1st Floor).

ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks. The issue was noted at LT Panel, Storage Area DB (Ground Floor), Cable Tray at 1st Floor Stitching Area, and Guard Room.

iii. Machine plugs were installed directly into incompatible sockets, creating unsafe electrical connections and increasing the risk of overheating, arcing, and fire. The issue was noted at 1st Floor Stitching Machines.

For any future electrical work, the termination of stranded cables must be done with cable lugs and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety.F-202603-064411

Timeline	Remediation Plan
Medium Term CAP-0075925	i. Install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022)
Medium Term CAP-0075926	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring.

Medium Term CAP-0075929	iii. It is recommended to avoid the use of unsafe plug-in arrangements and ensure that each machine is provided with a properly rated, dedicated socket—preferably of an industrial type. All electrical connections should be carried out in a safe and proper manner in accordance with good engineering practices, as required by BS 7671 (Regs. 553.1.7, 553.2.1, and 134.1.1).
Longer Term CAP-0075927	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.
Longer Term CAP-0075928	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Longer Term CAP-0075930	iii. Conduct regular safety audits and provide training sessions to staff to reinforce the importance of using proper power plugs and adherence to electrical safety standards.



Figure 2.49 - PVC tape joint at LT panel electrical room



Figure 2.50 - Lugs missing at cable terminations and PVC tape joints found at storage DB ground floor



Figure 2.51 - Lugs missing at cable terminations and PVC tape joints found at office DB first floor



Figure 2.52 - PVC tape joint at first floor stitching area cable tray



Figure 2.53 - Improper plug in arrangement for first floor stitching machines

11.3 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires. The issue was noted at cutting DB ground floor, LT panel and changeover panel at electrical room

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202603-064410

Timeline	Remediation Plan
Medium Term CAP-0075921	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0075922	Develop and enforce a strict protocol for cable colour coding to be followed in all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.54 - Color coding violation observed at change over panel electrical room



Figure 2.55 - Color coding violation observed at LT panel electrical room

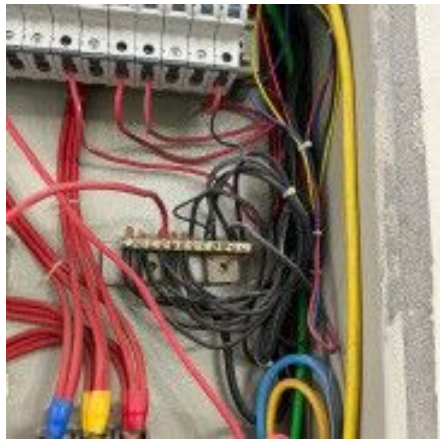


Figure 2.56 - Color coding violation observed at ground floor cutting hall DB

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.